

- Computer architecture
- Understanding peripherals, reference manuals, datasheets
- Protocols, networking, and moving data
- Programming languages and structure
- Validation and verification of electronic boards
- PCB designing
- Quality assurance and compliance
- Power and battery management

Different job types

In a broader sense, your expertise and skillset in embedded systems would depend on the type of your job—whether it is firmware or purely hardware, or both. “Good programming logic, if you work in firmware, and an extremely good electronics know-how as per professional need (such as PCB designing, layout, understanding NETLIST), if you work exclusively in hardware, is necessary,” says Balasubramanian.

Embedded hardware engineering: Embedded hardware engineers are in charge of all areas of electrical system design, including high-speed digital circuits, power management, PCB layouts, and thermal management. They work with the algorithm and software teams to design and build high-performance computing platforms. Also, testing the hardware is a critical part of embedded hardware development.

Embedded hardware systems are designed, implemented, and tested on a regular basis by hardware test engineers. They employ numerous testing methods and keep track of hardware modifications and hardware-related issues. They regularly test systems to eliminate potential faults and collaborate with cross-functional teams to give system-level support (mechanical, software, hardware, etc).

Embedded firmware engineering:

Firmware is a subset of software. It is often a program that controls an embedded system. Hence, firmware engineering includes both hardware and software. It might be part of a standalone product (code that is embedded into very small systems, such as a monitor, a mouse, or a washing machine) or it can run on a system that also runs higher-level software. For example, a laptop will contain a lot of firmware in the form of device drivers, etc.

While working closely with the hardware designers, a firmware engineer will focus mainly on the firmware. Firmware engineers create low-level software that can interact directly with the hardware, such as converting analogue sensor signals to digital data and managing communications protocols. We can say that firmware is a low-level interface to the hardware.

Embedded software engineering:

There is a lot of uncertainty and vagueness when it comes to differentiating between an embedded software engineer and a firmware engineer. Many new types of embedded devices have emerged in recent years—devices that are more complex than traditional embedded systems but less complex than personal computers. These utilise file systems on their flash to put programs into memory for execution, and they run larger operating systems like Linux and UNIX.

Embedded software programmers deal with these types of systems as well as old-style firmware on occasion. Routers, smart TVs, and even smartphones and tablets are among

the products they work on. Embedded software is basically software that is designed for a specific purpose and has a limited user interface. It will almost never be general-purpose, even if it is highly complicated.

Embedded systems engineering:

Embedded systems engineers can create both hardware and firmware, and deal with tradeoffs in order to achieve the highest performance at the lowest cost. On very small devices (of which there are many), we can see them mostly doing hardware and firmware design. Overall, they handle hardware design and then entire hardware-software integration.

Systems architecture is another main aspect that these engineers deal with on a regular basis. The design, development, production, testing, and maintenance of embedded systems are some of their responsibilities. “Simply put, embedded hardware engineers work on the hardware side, and embedded software engineers work on the software/firmware side. Embedded systems engineers work on both,” says Rosmianto Aji Saputro, senior embedded engineer at ZebraX.

Non-technical skills

Irrespective of your profession, you can never ignore soft skills. Some soft skills important for embedded system designers are:

- Communication skills
- Marketing
- User research
- Quality assurance
- Time management
- Problem-solving ability
- Ability to learn outside your comfort zone
- Creativity 

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